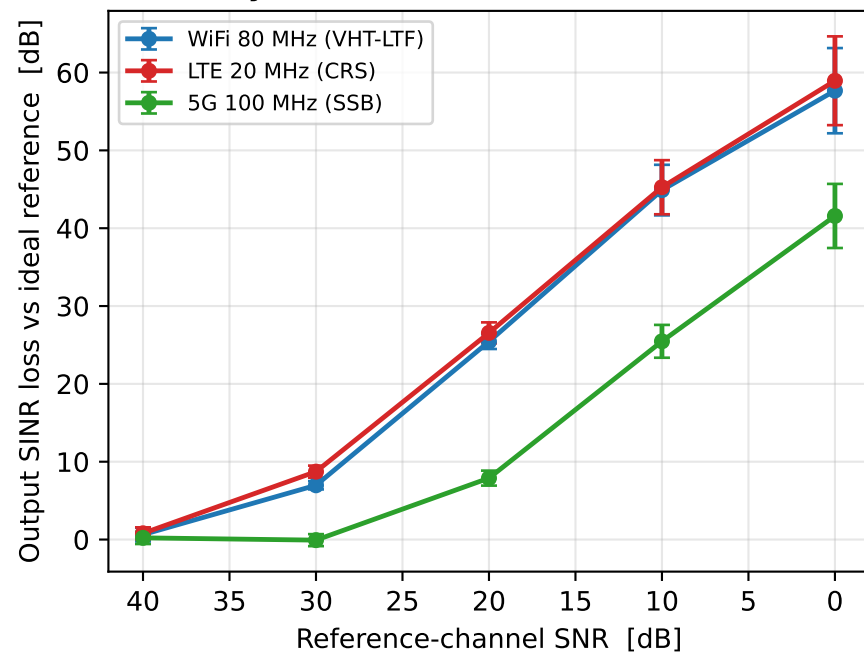
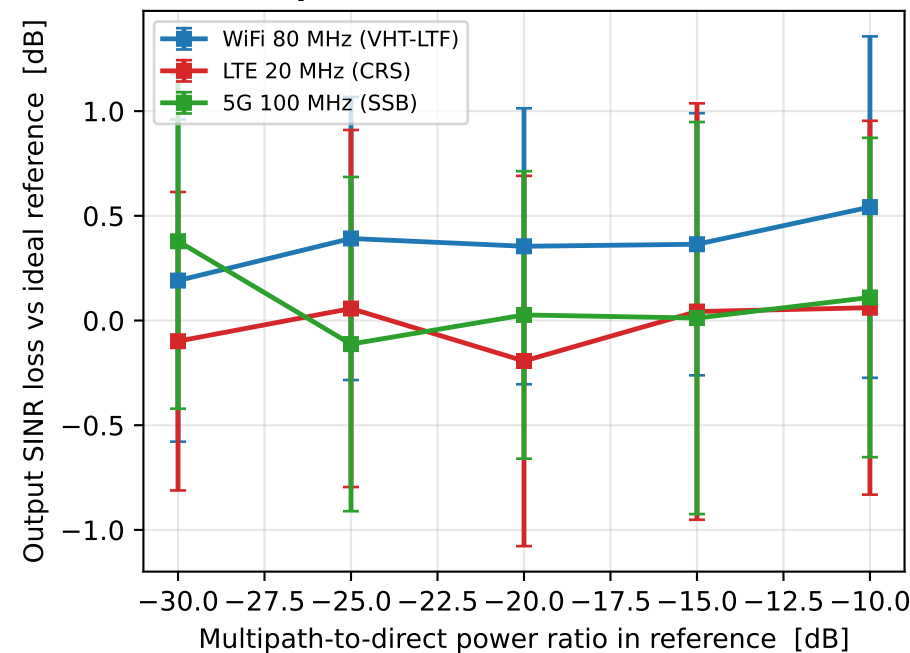


Passive radar with a real reference channel — what the two-channel promotion costs

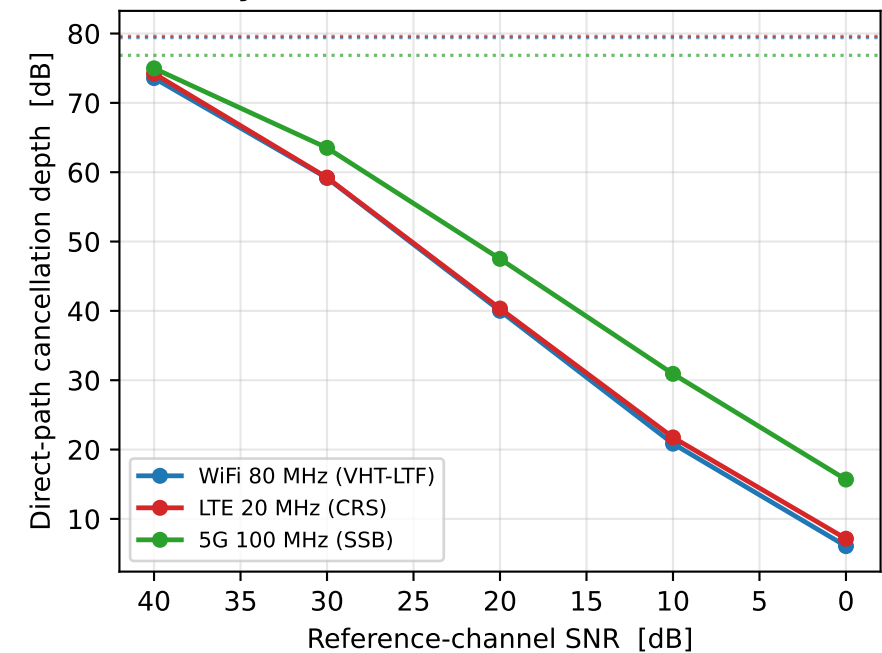
(a) Noisy reference channel



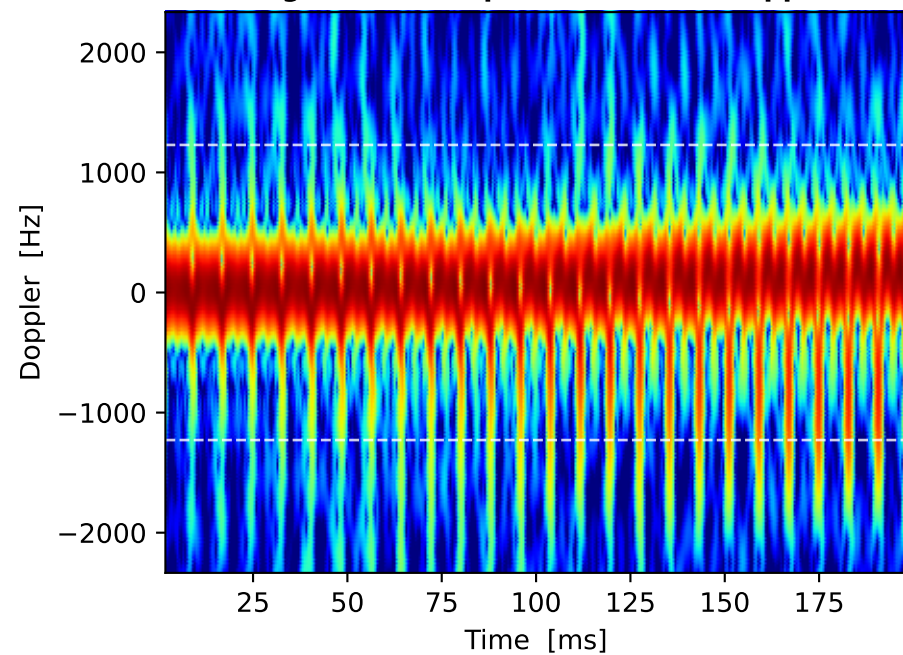
(b) Multipath-contaminated reference



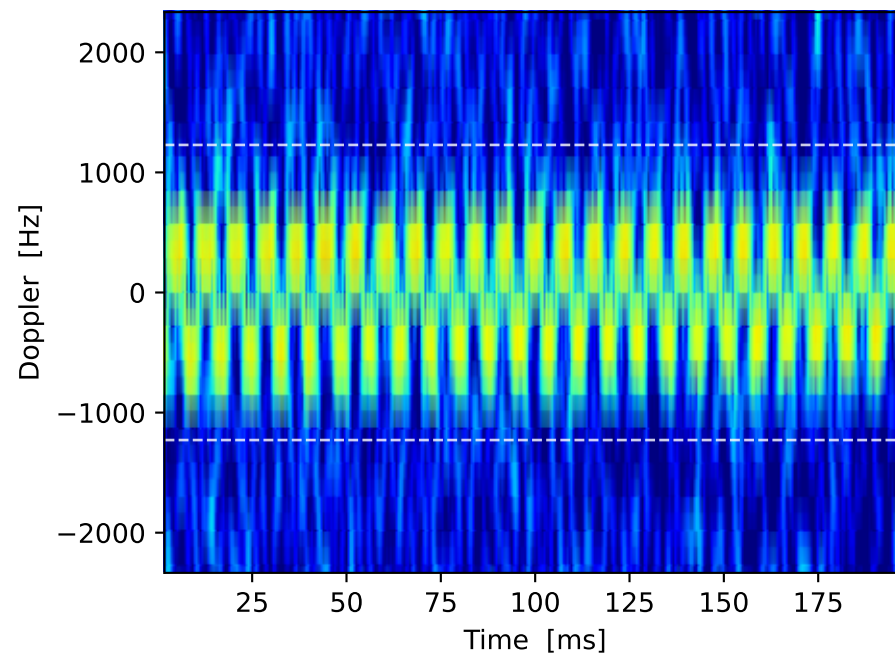
(c) Why it costs: ECA cannot cancel what it mismeasures



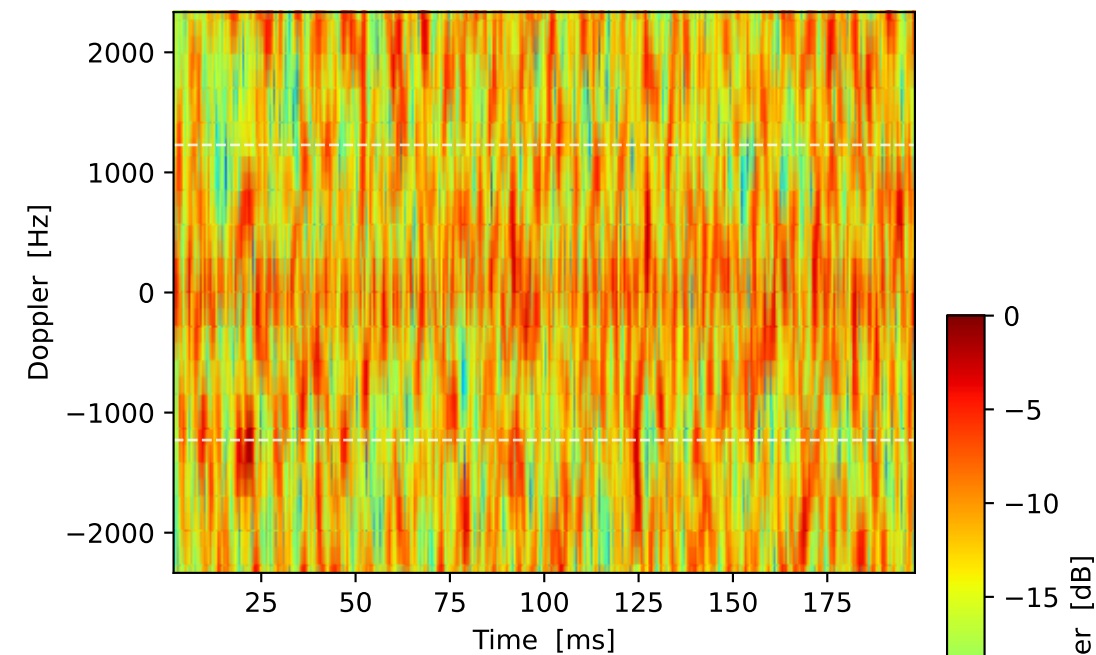
(d) Ledger STFT (no passive chain) — upper bound



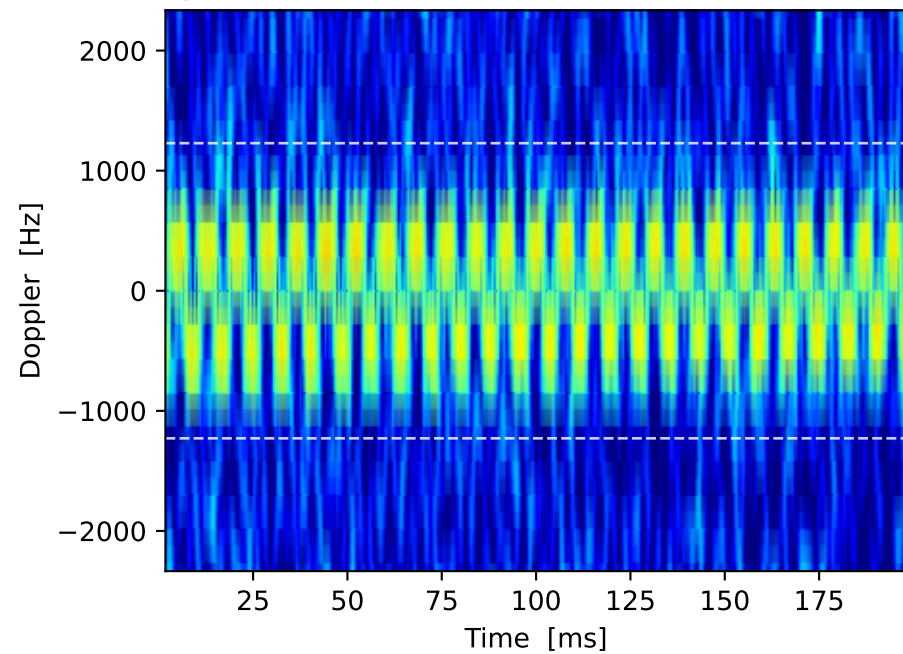
(e) Passive chain, ideal reference



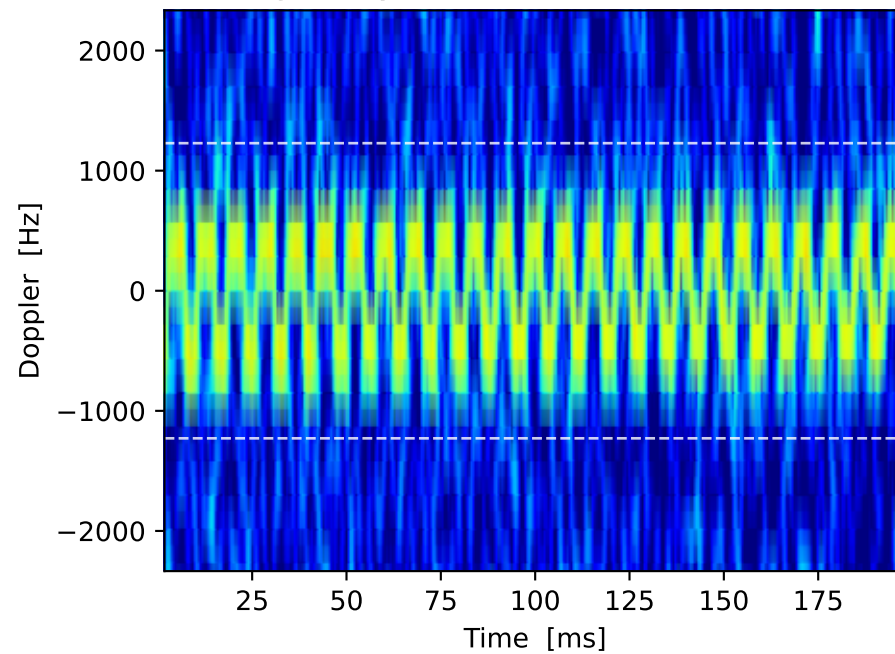
(f) Passive chain, reference SNR 10 dB



(g) Passive chain, multipath reference -20 dB



(h) Target only, ECA off — control for the zero-Doppler notch



(i) 5G frame rate 2 kHz — micro-Doppler aliases

